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QUESTION 1: ESSAY

Why is it important for chemisorption to occur for catalysis? Why not physical adsorption?

Given Answer: Chemisorption is important because it shows the energy absorbed from the molecules that are held to the surface by valence forces but physical only show the solid

Score: 25

QUESTION 2: ESSAY

List the seven steps of a catalytic reaction.

Given Answer: Mass Transfer IN
Pore Diffusion Into catalyst
Absorption of reactant
Reaction (Catalyst site)
Desorbtion
Pore Diffussion out
Mass Transfer OUT

Correct Answer: 
Bulk Flow
Mass-Transfer in
Internal Diffusion in
Adsorption
Reaction
Desorption
Diffusion out
Mass-Transfer out
Bulk Flow out

Score: 25

QUESTION 3: ESSAY

For $A \rightarrow B$ reaction (Pressure = P_A) occurring on a catalyst surface, the combined rate of adsorption, desorption and surface reaction steps is given by $-r_A = (P_A^3 - P_B^2/5)/(1 + P_A + P_B)^2$. The mass transfer coefficient for A is given by k_c . When the system is agitated, the overall rate of the process also changes, implying that this is a mass-transfer limited process and the surface reaction is occurring very fast. Find the overall rate of reaction (at bulk pressures = P_A and P_B). Assume no internal diffusion limitation.

Given Answer: Equals 1

SCORE: 0

QUESTION 4: ESSAY

What is the meaning of rate-limiting step in a catalytic reaction. Explain in terms of rate constants and resistances.

Given Answer: The rate limiting step does not allow the reaction to continue further into the process. (rate constants) $k_s \ll k_a, k_d$ or (resistances) $1/k_d \ll 1/k_a, 1/k_s$

Score: 25